LESSON PLAN · SCRATCH (GR 3–5) BEGINNER: SCRATCH GAME MAKERS

Lesson 9: Project: Catch the Apple

Module 3 | 50 minutes | Project: game play | Grades 3–5 · ages 8–10

Objective: Combine events, loops, conditions, coordinates and random numbers to build and test a playable catching game.

Success criteria

- My bowl moves with the arrow keys.
- My apple falls, can be caught, and restarts.
- I improved my game after a playtest.

Key vocabulary

- **game loop**: The forever loop that keeps a game running.
- **playtest**: Watching someone play to find problems.
- **difficulty**: How easy or hard a game is.
- **sprite interaction**: When one sprite reacts to touching another.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)
- Printed project brief

Timed lesson plan (50 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Catch game with a ball and bin.
5 min	**Watch & plan**	Project video; label the two rules on the brief.
25 min	**Build**	Drop-and-collect puzzle, then build the game in the Code Lab project.
8 min	**Playtest**	Swap and play; record observations.
7 min	**Improve & submit**	Change one thing; reflection; submit.

Unplugged warm-up

Catch game (5 min): one student drops a soft ball from different heights/places (random), another catches with a bin (bowl). Discuss: what makes it harder or easier?

Teaching the idea

The Module 3 project integrates events (keys), loops (forever), conditionals (touching, y < −170), coordinates and randomness into a playable game. Emphasise playtesting and balancing difficulty.

Analogy: The apple's forever loop is like a referee who watches every moment and blows the whistle when a rule happens.

Live demo: Live-build the apple's script; first forget the bottom rule (apple gets stuck), then add it. Ask students to suggest a speed and test it.

Questions to ask

Ask	Listen for
What are the apple's two rules?	Touching bowl → new apple; at bottom → new apple.
How can you make the game harder?	Faster fall, smaller bowl movement.
What did your playtester do that surprised you?	Any observation.

Differentiation

Support
Provide the bowl scripts pre-built; students build only the apple's brain with the printed card.

Stretch
Smooth movement (forever + if key pressed), a golden apple worth more, or apples that speed up.

Common misconceptions

Misconception	How to fix it
One if can check both rules	Each rule needs its own if (or an if-else chain).
The bowl needs the touching rule	Either sprite can check, but put it in one place — usually the apple.
Holding a key moves smoothly	Key-press hats run once per press; use forever + if key pressed for smooth movement.

Assessment

Project rubric (also on the project brief):

Criterion	1 · Beginning	2 · Developing	3 · Secure	4 · Excellent

Events & controls	Bowl doesn’t move	Moves one way only	Moves both ways smoothly enough to play	Smooth controls (forever + if key pressed) or extra controls
Loops & conditions	Apple doesn’t fall	Falls but rules don’t work	Falls forever with working catch and miss rules	Extra rules (golden apple, speed-up) that work
Coordinates & random	Fixed positions only	Random used but apple sometimes off-stage	Random x at the top every time	Uses coordinates cleverly (difficulty, multiple apples)
Game design	Hard to play	Playable but unbalanced	Fun and fair	Polished: sounds, backdrop, instructions, balanced difficulty
Testing & reflection	No playtest	Played it alone	Playtested; reflection explains a change	Several improvements from testing explained clearly

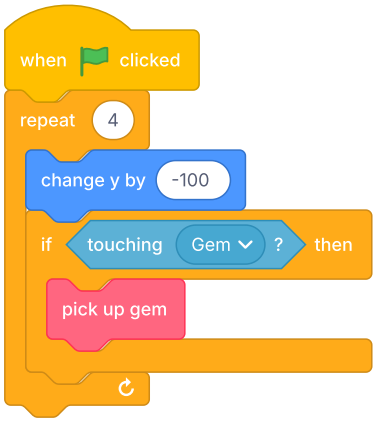
Exit ticket
Write the apple’s two if rules in words.

Homework
Teach someone at home to play your game. Ask them for one improvement idea.

Answer key — Kitabash activities

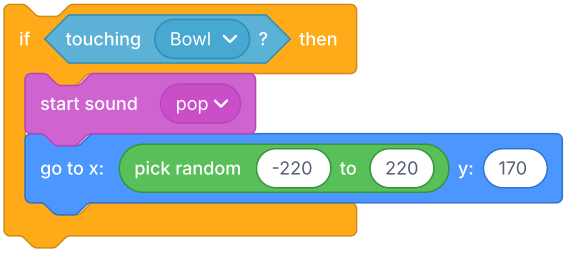
1. Drop and collect (perform · block code)

One solution:



2. Find the catch rule (perform · multiple choice)

Answer: A



3. Project: Catch the Apple (probe · code lab)

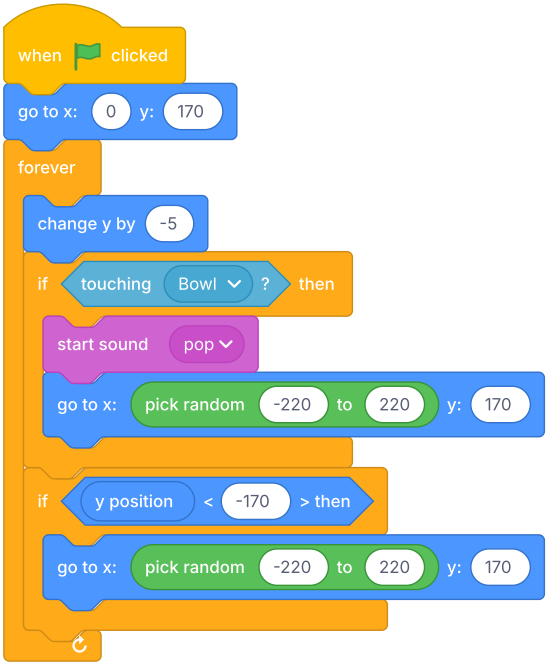
Goals checked: Bowl moves right · Bowl moves left · Apple falls · Catch rule · Catching plays a sound

Example project:

Bowl



Apple



4. Explain your thinking (probe · free response)

Look for: played, change, because. **Example:** My friend played and missed lots of apples because the bowl was too slow. I will change the bowl to move 30 steps so the game is fairer.

Answer key — worksheet

1. **Sprites and scripts:** Bowl: start position + arrow keys. Apple: falls forever, catch rule, miss rule.
2. **Rules:** Play a sound and go back to a random spot at the top · go back to a random spot at the top.

3. **Playtest notes:** Observations.
4. **My improvement:** A justified change (speed, size, sound…).

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.